

Yangfan Deng

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EDUCATION

- **University of Maryland, College Park, College Park, MD, United States** 2025 - present
Ph.D. student in Electrical and Computer Engineering, A. James Clark School of Engineering
Advisor: Dr. Min Wu Dean's Fellowship
- **University of Pittsburgh, Pittsburgh, PA, United States** 2023 - 2025
M.S. in Electrical and Computer Engineering, Swanson School of Engineering GPA: 3.93/4.00
- **Ocean University of China, Qingdao, Shandong, China** 2018 - 2022
B.S. in Computational Mathematics, School of Mathematical Sciences GPA: 3.52/4.00 (Top 10%)

PROFESSIONAL EXPERIENCE

- **IBM, AI Research Intern, Shenzhen, China** Jun. 2021 - Sept. 2021
 - Led a three-person intern team in Pressfit project to design algorithm and display results.
 - Designed a deep learning based algorithm to predict the deformation force during the Pressfit process.
 - Utilized IBM cloud, DB2 to establish a front-end platform to display the curve of press force.
 - Presented the internship defense to the Global Vice President of IBM Supply Chain and received the title of Outstanding Intern.
- **China Mobile, Assistant Project Manager, Wuhan, China** Jan. 2019 - Feb. 2019
 - Assisted the project manager in managing and ensuring the smooth operation of all departments involved in the project.
 - Participated in infrastructure construction of Wuhan Internet of Things Project, and promoted the adoption of smart home technology and healthcare solutions.
 - Contributed in the popularization of the base station in Hongshan District in Wuhan, and participated in deciding the layout of base station signal transmitters.

RESEARCH EXPERIENCE

Scenario-Driven and Keyed Watermarking Framework for 3D Gaussian Splatting

Advisor: Professor Min Wu, University of Maryland, College Park

Sep. 2025 - now

- Proposed a scenario-driven threat-model blueprint for 3DGS watermarking and evaluated its spread-spectrum baseline.
- Analyzed practical deployment scenarios for 3DGS watermarking and distilled a systematic set of system requirements.
- Developing a keyed 3DGS watermarking framework for stronger security guarantees, targeting major security gaps in existing 3DGS watermarking pipelines.
- One conference paper is currently under review.

Topological and Interpretable Dimensionality Reduction Techniques for EEG Signal Analysis

Advisor: Professor Zhi-Hong Mao, University of Pittsburgh

Sep. 2023 - Apr. 2025

- **Topological and Spectro-temporal Analysis for Dimensionality Reduction Using EEG Signal**
 - Conducted Topological Data Analysis on EEG signals corresponding to different sleep stages to validate the effectiveness of topological features.
 - Designed an advanced analysis of EEG signals by extracting features using a combination of topological and spectro-temporal analysis techniques, providing feature candidates for dimensionality reduction algorithms.
 - Proposed a two-stage feature selection framework to enhance the performance of explainable dimensionality reduction algorithms.
 - The journal paper resulting from this project is currently under review by **ACM HEALTH**.
- **Interpretable Dimensionality Reduction Algorithms Using EEG Signal**
 - Applied manifold learning based algorithms, such as UMAP, to EEG signals.
 - Achieved 2D visualization of EEG data and performed mathematical analysis and derivation of clustering results.
 - Led the progress of the project and efficiently collaborated with group members with different technical backgrounds.
 - Collaborated with Professor Mao to guide high school student David Zhang in completing his research project.
 - One conference paper resulting from this project has been accepted by **EMBC 2024**. Another conference paper has been accepted by **ICDSCA 2024**.

Innovative Approaches to Object Grasping and Recognition Based on Deep Learning

Advisor: Professor Yong Zhao, Ocean University of China

Jun. 2021 - May. 2024

- **Object Grasping Algorithm under Limited Condition**
 - Proposed two loss functions for grasping algorithms based on inaccurate or missing ground truth in training data, filling a gap in the field.
 - Enhanced the robustness of the grasping network under different conditions of missing and inaccurate training data.

- The conference paper resulting from this project has been posted on arXiv (planned submission for **CASA 2025**).
- **Lightweight Object Grasping Algorithm**
 - Proposed a new lightweight neural network for object grasping based on GhostNet, filling a gap in the field at that time.
 - Generate 2D grasp visualizations on RGB-D images, achieving a 94% grasp success rate.
 - The conference paper resulting from this project has been accepted by **FGIP 2022**.
- **Smile Recognition Based on Bayesian Methods**
 - Deployed Python web crawler to construct smile comprehensive dataset.
 - Cooperated to design the routine of the Bayesian classifier for smile recognition.
 - The conference paper resulting from this project has been accepted by **ICIGP 2023**.

PUBLICATIONS

- [1] **Yangfan Deng**, Anirudh Nakra, and Min Wu, “3D Gaussian Splatting Watermarking Should Be Scenario-Driven and Threat-Model Explicit”, *currently under review*. arXiv: 2602.02602
- [2] **Yangfan Deng**, Hamad Albidah, Ahmed Dallal, Jijun Yin, and Zhi-Hong Mao, “Two-Stage Hierarchical and Explainable Feature Selection Framework for Dimensionality Reduction in Sleep Staging”, **IEEE Transactions on Biomedical Engineering**, *currently under review*. arXiv:2409.00565.
- [3] **Yangfan Deng**, Hamad Albidah, Haoliang Cheng, Ahmed Dallal, Jijun Yin, and Zhi-Hong Mao, “UMAP for Dimensionality Reduction in Sleep Stage Classification Using EEG Data”, **Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2024)**, Orlando, Florida, USA, *accepted for publication*, July 2024.
- [4] **Yangfan Deng**, Lulu Wu, and Yong Zhao, “Robust Loss Functions for Object Grasping under Limited Ground Truth”. arXiv: 2409.05742.
- [5] Ariel Yin, David Zhang, David Mao, Sichuang Li, Haoliang Cheng, **Yangfan Deng**, Yifan Guo, Helen Mao, Jijun Yin, and Zhi-Hong Mao, “Design concept of a wearable device for sleep related brain wave detection and stimulation”, **Proceedings of the 4th IEEE International Conference on Data Science and Computer Application (ICDSCA 2024)**, Dalian, China, *accepted for publication*, October 2024.
- [6] Junjie Huang, **Yangfan Deng**, Qinghua Guo, Yizhou Xu, Qingtao Pan, and Yong Zhao, “Smile Recognition Based on Comprehensive Dataset Construction and Bayesian Neural Architecture Search”, **Proceedings of 6th International Conference on Image and Graphics Processing (ICIGP 2023)**, Chongqing, China, *accepted for publication*, January 2023.
- [7] **Yangfan Deng**, Qinghua Guo, Yong Zhao, and Junji Xu, “A Lightweight Object Grasping Network using GhostNet”, **Proceedings of 2nd International Workshop on Frontiers of Graphics and Image Processing (FGIP 2022)**, Beijing, China, *accepted for publication*, November 2022.

HONORS

- Contemporary Undergraduate Mathematical Contest in Modeling (MCM): **Won the National Second Prize**, 2021
- MathorCup University Mathematical Modeling Challenge: **Won the Third Prize**, 2020
- Contemporary Undergraduate Mathematical Contest in Modeling (MCM): **Won the Second Prize in Shandong Province**, 2019
- First-Class Academic Scholarship: **GPA Top 5%**, 2019
- Mathematical Contest in Shandong Province: **Won the Third Prize**, 2019

PROFESSIONAL SKILLS

- Programming language: Python, C++, MATLAB, SQL.
- AI frameworks: PyTorch and TensorFlow.
- Language: Professional in English.